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The Margins of Global Sourcing: Theory and Evidence from US Firms

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1 Big picture

- **Question.** How do firms choose the set of countries from which to source intermediate inputs, and how do these extensive-margin sourcing decisions shape firm size, input purchases, domestic sourcing, and aggregate trade responses to shocks?
- **Core problem.** Importing is not like exporting in canonical models. In many export models, entry into destination markets is additively separable because marginal cost is fixed. In sourcing, however, the purpose of importing is precisely to lower marginal cost. Therefore, entering one source country changes the marginal value of entering another country.
- **Main theoretical insight.** Global sourcing decisions can be complements or substitutes across countries. Under the empirically relevant case, in which the demand elasticity is high relative to input-productivity dispersion, adding one source country raises the value of adding other source countries.
- **Computational challenge.** If there are J possible source countries, a firm formally chooses among 2^J country bundles. With 66 foreign countries, brute force enumeration is infeasible.
- **Solution method.** The paper exploits complementarity and adapts Jia’s algorithm for discrete choices with strategic complementarities to solve firms’ sourcing sets in a high-dimensional environment.
- **Data.** The paper uses confidential US Census firm-level import and production data, with source-country information, to study US manufacturing firms and their importing decisions in many countries.
- **Structural estimate.** The baseline estimate implies that a firm sourcing from all foreign countries faces roughly 9 percent lower variable costs and achieves roughly 32 percent higher sales than if it sourced only domestically.
- **Counterfactual.** A China sourcing-potential shock calibrated to match the growth of Chinese input sourcing between 1997 and 2007 produces heterogeneous responses across firms. New China importers expand domestic and third-market sourcing, while non-China importers shrink.
- **Reduced-form validation.** Using a firm-level China sourcing shock, the paper shows that exogenous increases in Chinese sourcing are associated with increased domestic and third-market sourcing, not simple substitution away from other sources.
- **Economic takeaway.** Global sourcing shocks do not just reallocate input purchases from domestic suppliers to foreign suppliers. Because sourcing lowers marginal costs and changes firm scale, the same shock can make some firms expand all margins while making other firms contract.

2 Incremental contribution

- **From export entry to import sourcing.** The paper shifts the standard heterogeneous-firm trade framework from export-market entry to import-source entry. This is not a cosmetic change: sourcing decisions affect the firm’s cost function, so market-entry decisions become interdependent.
- **Multi-country extensive margin.** Earlier theory often handled one or two source markets, or abstracted from fixed sourcing costs. This paper gives a tractable characterization of firms’ extensive-margin sourcing decisions across many countries.
- **Complementarity as a solution device.** The paper shows that a simple parameter condition transforms the source-country choice problem into a complementarity problem that can be solved without searching all 2^J bundles.
- **Quantitative global-sourcing model.** The paper combines Eaton-Kortum input technology with firm heterogeneity and fixed sourcing costs, yielding a model that can be estimated using firm-level sourcing patterns and country characteristics.

- **Heterogeneous firm-country fixed costs.** A major empirical insight is that country-level attractiveness is not enough. Firms have idiosyncratic fixed costs of sourcing from each country, which helps the model match the observed heterogeneity in importing strings.
- **New counterfactual margin.** The paper shows that a China sourcing shock affects not only sourcing from China, but also sourcing from the United States and from third countries. These third-market effects are the key empirical signature of interdependent sourcing decisions.
- **Gross versus net sourcing effects.** The paper emphasizes that aggregate net changes can hide large gross changes. Some firms increase domestic sourcing after the China shock even while aggregate domestic sourcing falls.
- **Methodological spillover.** The algorithmic approach is useful beyond this paper: it provides a way to solve high-dimensional firm choices with complementarities in other trade, IO, and organizational settings.

3 Data and measurement

3.1 Firm-level sourcing data

- The paper uses confidential US Census data linking US manufacturing firms to import transactions by product and source country.
- The baseline descriptive and structural evidence focuses on 2007, a year with rich information on firm sales, input purchases, and import sourcing.
- The counterfactual validation uses changes from 1997 to 2007, a period spanning China’s accession to the WTO and a large increase in US sourcing from China.
- A source country is active for a firm if the firm imports from that country. Domestic sourcing is measured through domestic input purchases.

Interpretation: The data allow the paper to observe the extensive margin of importing directly: not merely whether a firm imports, but from how many countries and which countries.

3.2 Source countries and sourcing potential

- Source countries differ in wages, distance to the United States, language, institutions, R&D, population, and other characteristics.
- The paper estimates country sourcing potential from firms’ observed import shares, using a specification consistent with the model’s input-choice structure.
- The ranking of countries by number of importers differs from the ranking by import value, motivating the distinction between fixed costs and marginal sourcing potential.

Interpretation: Countries can be attractive because they offer low variable input costs, low fixed sourcing costs, or both. The data show that these two dimensions are not the same.

3.3 Fixed sourcing costs

- The structural model allows firm-country-specific fixed costs of sourcing.
- Fixed costs depend on distance, common language, institutional quality, and a firm-country idiosyncratic component.
- Domestic fixed sourcing costs are normalized to zero because active US manufacturers must use domestic inputs.

Interpretation: Firm-country heterogeneity is crucial. If all firms faced the same country ordering, the model would generate too rigid a pecking order and would fail to match the breadth of observed sourcing patterns.

3.4 China shock measures

- Structurally, the paper calibrates a China sourcing-potential shock to match the observed increase in Chinese import shares from 1997 to 2007.
- Reduced-form evidence uses a firm-level China shock in the spirit of shift-share designs: firms are differentially exposed depending on the input industries they use and China-related changes in those input markets.
- The outcomes include domestic input purchases, foreign input purchases, number of source countries, and firm employment.

Interpretation: The empirical validation asks whether firms more exposed to the China sourcing opportunity grow domestic and third-country sourcing, as the complementarity model predicts.

4 Models and theoretical specifications

4.1 Production and sourcing capability

Firms produce differentiated final goods using a continuum of intermediate inputs. A firm with core productivity φ can source each input from the cheapest active country in its sourcing set $\mathcal{J}$.

A key sufficient statistic is the sourcing capability of the set $\mathcal{J}$:

$$\Theta_i(\mathcal{J}) = \sum_{j \in \mathcal{J}} T_j (w_j \tau_{ij})^{-\theta},$$

where T_j captures country technology, w_j is the wage, τ_{ij} is the iceberg sourcing cost, and θ controls dispersion in input efficiency draws.

Interpretation: A larger $\Theta_i(\mathcal{J})$ means the firm can draw cheaper inputs from its set of source countries. Because the firm picks the lowest-cost source for each input, adding a country increases the option value of the whole set.

4.2 Variable profits and fixed sourcing costs

The firm's operating profits from sourcing set $\mathcal{J}$ can be written in reduced form as

$$\pi_i(\varphi, \mathcal{J}) = B_i \varphi^{\sigma-1} [\Theta_i(\mathcal{J})]^{(\sigma-1)/\theta} - \sum_{j \in \mathcal{J}} f_{ij},$$

where σ is the final-good demand elasticity and f_{ij} is the fixed cost of sourcing from country j .

Interpretation: The first term captures the revenue and variable-cost benefit of sourcing capability. The second term captures the fixed costs of maintaining source-country relationships. The firm's problem is an extensive-margin choice over a bundle of source countries.

4.3 Complementarity versus substitutability

The curvature of

$$[\Theta_i(\mathcal{J})]^{(\sigma-1)/\theta}$$

controls whether source countries are complements or substitutes. When

$$\frac{\sigma-1}{\theta} > 1,$$

adding one country raises the marginal value of adding another country.

Interpretation: This is the empirically important complementarity case. If sourcing from one country lowers marginal cost and increases firm scale, then the value of adding a second country can rise rather than fall.

4.4 Firm problem

The firm chooses a sourcing set:

$$\mathcal{J}^*(\varphi) = \arg \max_{\mathcal{J} \subseteq \mathcal{N}} \left\{ B_i \varphi^{\sigma-1} [\Theta_i(\mathcal{J})]^{(\sigma-1)/\theta} - \sum_{j \in \mathcal{J}} f_{ij} \right\}.$$

Interpretation: In a 66-country setting, this is a massive combinatorial problem. The theoretical complementarity result allows the authors to solve it by iterating on upper and lower bounds rather than enumerating every possible sourcing set.

4.5 Marginal benefit of adding a country

For a candidate sourcing set $\mathcal{J}$ and country $j \notin \mathcal{J}$, the marginal benefit of adding j is

$$MB_j(\mathcal{J}) = B_i \varphi^{\sigma-1} \left(\Theta_i(\mathcal{J} \cup j)^{(\sigma-1)/\theta} - \Theta_i(\mathcal{J})^{(\sigma-1)/\theta} \right) - f_{ij}.$$

Interpretation: If $MB_j(\mathcal{J}) > 0$, adding country j is profitable at the current set. Under complementarity, adding other countries tends to increase these marginal benefits.

4.6 Trade-flow and gravity interpretation

The model nests familiar gravity structures as special cases. With no fixed costs, all firms import from all locations and the model resembles Eaton-Kortum gravity. In knife-edge cases with independent entry, it resembles standard extensive-margin trade models.

Interpretation: The general model is richer because entry into one source country affects entry incentives elsewhere. This is the central departure from additive export-market entry models.

5 Estimation and empirical specifications

5.1 Sourcing potential regression

The paper estimates country sourcing potential using firm-country import shares. In reduced form, the estimated country terms are related to country characteristics such as wages, distance, R&D, capital per worker, common language, corruption control, and the number of firms.

Interpretation: This step identifies the variable-cost attractiveness of source countries from the pattern of firm sourcing shares.

5.2 Trade elasticity estimation

The paper estimates firm-level and aggregate trade elasticities using country-level and firm-level sourcing information, with instruments such as population and human-capital-adjusted wage variation.

Interpretation: The elasticity determines how strongly sourcing shares respond to cost differences. It is also central for deciding whether sourcing decisions are complements.

5.3 Simulated method of moments

The fixed cost parameters are estimated by matching model-generated moments to observed sourcing patterns:

- the share of firms that import;
- import participation by firm size;
- the share of importing firms by country;
- hierarchical sourcing patterns;
- aggregate sourcing shares.

Interpretation: The estimation uses both extensive-margin and aggregate moments, so the model must match who imports, from where, and how large the flows are.

5.4 Counterfactual China sourcing shock

The paper changes China’s sourcing potential so that the model matches the growth of China’s import share between 1997 and 2007. It then compares outcomes under the baseline model to alternatives with fixed sourcing strategies, independent entry decisions, universal importing, or common fixed costs.

Interpretation: The counterfactual isolates the role of interdependencies. If extensive-margin complementarities matter, a China shock should also change domestic and third-country sourcing.

5.5 Reduced-form China shock regression

The paper estimates regressions of firm-level sourcing changes on a China sourcing shock:

$$\Delta Y_{i,1997-2007} = \alpha + \beta \cdot ChinaShock_i + \varepsilon_i,$$

where Y_i includes domestic inputs, foreign inputs, number of countries, and employment. The IV version instruments firm-level Chinese sourcing growth with a shock built from China market-share changes in EU15 countries.

Interpretation: The reduced-form design checks whether observed data move in the same qualitative direction as the model’s counterfactual predictions.

6 Identification and empirical design

6.1 What is identified structurally?

The structural estimation identifies the distribution of firm-country sourcing fixed costs and country sourcing potentials by fitting the model to detailed firm-level sourcing patterns.

Interpretation: The model is identified not by one regression coefficient, but by the joint pattern of country entry, import shares, and firm-size selection.

6.2 Why country-source rankings matter

The data show that Canada is the most common source country by number of importers, while other countries such as China and Mexico rank differently by import value. This is useful because it separates fixed costs from variable-cost potential.

Interpretation: A country can be easy for many firms to enter but not dominate import values, or it can generate high import values for fewer firms. The model needs both dimensions.

6.3 Why hierarchy tests matter

If firms faced common country-level fixed costs, they would tend to follow a strict pecking order. The data display some hierarchy, but not enough to be explained by common fixed costs alone.

Interpretation: The hierarchy evidence motivates the inclusion of firm-country fixed-cost heterogeneity.

6.4 Why reduced-form validation matters

The structural model predicts that a positive China sourcing shock can expand other sourcing margins. The reduced-form analysis checks whether firms exposed to the China shock actually increase domestic and third-country sourcing.

Interpretation: This provides external validation for the mechanism of sourcing complementarities.

6.5 Threats and limitations

- The model abstracts from some input-level and supplier-level fixed costs, though the online appendix discusses extensions.
- The China shock may be correlated with firm demand or output shocks, though the paper uses instruments and robustness checks to mitigate this concern.

- The model treats firms' country sourcing choices as the central extensive margin and does not fully model input-level contracting or intrafirm versus arm's-length sourcing.
- The counterfactual focuses on manufacturing sourcing and should not be read as a complete model of all labor-market effects of China trade.

Interpretation: The evidence is strongest for the mechanism that sourcing decisions are interdependent. The exact magnitudes depend on model structure and data measurement.

7 Section-by-section study guide

- **Introduction.** Motivates why importing is different from exporting and why source-country choices interact through marginal costs.
- **Theory.** Builds a multi-country sourcing model with Eaton-Kortum input technology, firm heterogeneity, and country fixed costs.
- **Data and descriptive evidence.** Shows that firms import from multiple countries, often follow partial hierarchies, and display sales premia increasing with sourcing breadth.
- **Estimation.** Estimates country sourcing potential, trade elasticity, and fixed cost parameters using firm-level sourcing data.
- **Counterfactuals.** Applies the model to a China sourcing-potential shock and compares baseline results to models that shut down key mechanisms.
- **Reduced-form evidence.** Tests whether actual firm-level sourcing changes from 1997 to 2007 are consistent with model predictions.
- **Conclusion.** Emphasizes that extensive-margin sourcing interdependencies generate heterogeneous firm responses to trade shocks.

8 Tables and figures

8.1 Figure 1: Sales Premia and Minimum Number of Sourcing Countries in 2007 + Table 1: Top Ten Source Countries for US Firms, by Number of Firms

Read:

- Figure 1 plots the sales premium associated with sourcing from at least a given number of countries.
- The premium rises almost monotonically with the minimum number of source countries.
- Firms that source from many countries are much larger than firms that source from few countries.
- Table 1 shows that countries differ sharply in number of importers and value of imports.
- Canada is the most common source country by number of firms, while China and Mexico are especially important by import value.

Detailed interpretation: The figure documents strong selection: broader global sourcing is associated with much larger firm size. The table shows that country attractiveness has at least two dimensions. Some countries are widely used by many firms, while others generate large import values among fewer firms. This combination motivates a model with both variable sourcing potential and fixed costs of sourcing.

Economic message: The extensive margin of sourcing is economically important. It is not enough to model total import intensity; one must model which countries enter the firm's sourcing set.

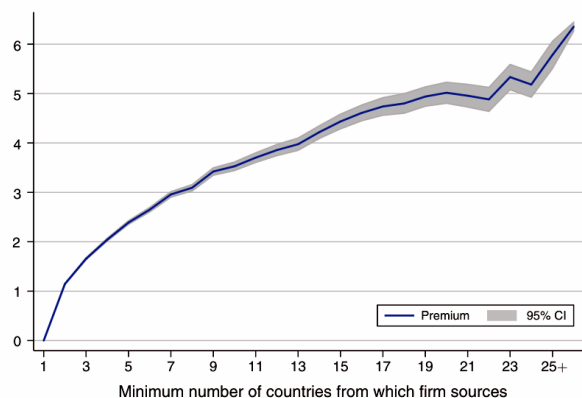


Figure 1: Sales Premia and Minimum Number of Sourcing Countries in 2007

	Rank by		Number of importers		Value of imports	
	Firms	Value	Firms	Percent of total	Imports	Percent of total
Canada	1	1	37,810	59	145,740	16
China	2	3	21,460	33	121,990	13
Germany	3	5	13,050	20	62,930	7
United Kingdom	4	6	11,450	18	30,750	3
Taiwan	5	11	10,490	16	16,630	2
Italy	6	13	8,500	13	13,230	1
Japan	7	4	7,980	12	112,250	12
Mexico	8	2	7,810	12	125,980	14
France	9	9	6,110	9	22,980	3
Korea, South	10	10	5,610	9	20,390	2

Table 1: Top Ten Source Countries for US Firms, by Number of Firms

8.2 Table 2: Firm-Level Statistics on the Number of Source Countries per Imported Product + Table 3: US Firms Importing from Strings of Top Ten Countries

Read:

- Table 2 shows that the typical imported product is sourced from a small number of countries.
- The median firm-product is sourced from one country, consistent with the model's lowest-cost-source logic.

- Table 3 documents that many firms follow partial hierarchical sourcing patterns among the top ten countries.
- About 36 percent of importers follow the pecking-order string in the data, compared with about 20 percent under random entry.

Detailed interpretation: Table 2 validates the idea that a firm can source an input from a single preferred location. Table 3 validates the idea that some countries are generally better sources than others. But the pecking order is incomplete, indicating that country-level fixed costs alone cannot explain sourcing choices. Firm-country idiosyncratic fixed costs are needed.

Economic message: Firms behave partly as if countries have a common attractiveness ranking, but not perfectly. This is exactly the empirical pattern the model is designed to rationalize.

TABLE 2—FIRM-LEVEL STATISTICS ON THE NUMBER OF SOURCE COUNTRIES PER IMPORTED PRODUCT

	Firm level		
	Mean	Median	Max
Mean	1.11	1.03	1.78
Median	1.00	1.00	1.00
95th percentile	1.61	1.00	4.00

Notes: Table reports statistics on the number of countries from which a firm imports the same HS10 product.

Table 2: Firm-Level Statistics on the Number of Source Countries per Imported Product

TABLE 3—US FIRMS IMPORTING FROM STRINGS OF TOP TEN COUNTRIES

String	Data		Random entry	
	Firms	Percent of importers	Firms	Percent of importers
CA	17,980	29.82	6,760	11.21
CA-CH	2,210	3.67	3,730	6.19
CA-CH-DE	340	0.56	1,030	1.71
CA-CH-DE-GB	150	0.25	240	0.40
CA-CH-DE-GB-TW	80	0.13	50	0.08
CA-CH-DE-GB-TW-IT	30	0.05	10	0.02
CA-CH-DE-GB-TW-IT-JP	30	0.05	0	0.00
CA-CH-DE-GB-TW-IT-JP-MX	50	0.08	0	0.00
CA-CH-DE-GB-TW-IT-JP-MX-FR	160	0.27	0	0.00
CA-CH-DE-GB-TW-IT-JP-MX-FR-KR	650	1.08	0	0.00
TOTAL following pecking order	21,680	36.0	11,820	19.6

Notes: The string CA means importing from Canada but no other among the top ten; CA-CH means importing from Canada and China but no other; and so forth. Percent of importers shows percent of each category relative to all firms that import from top ten countries.

Table 3: US Firms Importing from Strings of Top Ten Countries

8.3 Figure 2: Country Sourcing Potential Parameters and the Extensive Margin + Table 4: Estimation of Firm and Aggregate Trade Elasticities

Read:

- Figure 2 relates estimated country sourcing potential to the number of US importers from that country.
- The strong positive relation shows that countries estimated to have higher sourcing potential are used by more firms.
- Table 4 estimates firm-level and aggregate trade elasticities.
- The preferred IV firm-level elasticity is around 1.789, which is close to the value needed for sourcing complementarities given the chosen demand elasticity.

Detailed interpretation: Figure 2 shows that the estimated country fixed effects have economic content: they line up with the extensive margin of importer participation. Table 4 gives the key elasticity discipline for the model. The trade elasticity determines how much sourcing responds to cost differences and enters the complementarity condition.

Economic message: The estimates give empirical support to the paper's key modeling choice: source countries are not independent options; they interact through the firm's marginal cost and scale.

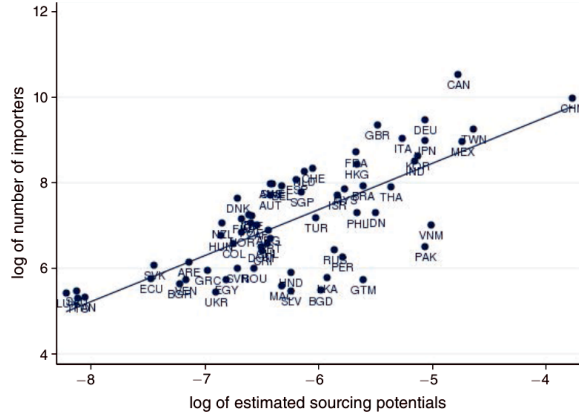


FIGURE 2. COUNTRY SOURCING POTENTIAL PARAMETERS AND THE EXTENSIVE MARGIN

Figure 2: Country Sourcing Potential Parameters and the Extensive Margin

TABLE 4—ESTIMATION OF FIRM AND AGGREGATE TRADE ELASTICITIES

	log ζ		log aggregate imports		
	OLS	IV	OLS	IV	IV
log HC adjusted wage	-0.537 (0.184)	-1.789 (0.696)	-0.643 (0.390)	-4.544 (1.844)	-1.268 (0.768)
log distance	-0.341 (0.197)	-0.621 (0.294)	-0.859 (0.418)	-1.733 (0.779)	-0.650 (0.333)
log R&D	0.352 (0.068)	0.524 (0.125)	0.763 (0.144)	1.298 (0.332)	0.251 (0.176)
log capital/worker	-0.184 (0.175)	0.425 (0.390)	-0.264 (0.370)	1.633 (1.033)	0.308 (0.421)
Common language	0.105 (0.223)	0.146 (0.289)	0.354 (0.471)	0.479 (0.764)	0.137 (0.317)
Control corrupt	0.156 (0.151)	0.621 (0.312)	0.365 (0.319)	1.816 (0.826)	0.414 (0.350)
log number of firms	0.108 (0.086)	-0.020 (0.130)	0.031 (0.183)	-0.369 (0.345)	-0.062 (0.142)
log domestic purchases					2.392 (0.327)
Constant	-7.250 (0.922)	-11.068 (2.323)	14.499 (1.952)	2.600 (6.156)	-37.389 (6.573)
Observations	57	57	57	57	57

Notes: Standard errors in parentheses. In the IV specifications, the human-capital-adjusted wage is instrumented by population. HC adjusted wage is country wage adjusted for differences in human capital. Domestic purchases is total purchases of US inputs by firms sourcing from a country. First-stage F -statistic on the excluded instrument is 6.49. First-stage regression results are in the Appendix.

Table 4: Estimation of Firm and Aggregate Trade Elasticities

8.4 Table 5: Estimated Parameters + Figure 3: Estimated Sourcing Potential and Median Fixed Costs by Country

Read:

- Table 5 reports the estimated fixed-cost parameters.
- Distance increases fixed costs, common language lowers fixed costs, and control of corruption affects sourcing costs.
- Figure 3 shows country-level sourcing potential against median fixed costs.
- Canada combines high sourcing potential with very low median fixed cost; China has high sourcing potential but higher fixed costs.

Detailed interpretation: Table 5 confirms that fixed costs are economically related to gravity variables and institutional variables. Figure 3 shows why the top-source rankings are not one-dimensional: countries can be high-potential but costly to enter, or lower-potential but easy to enter.

Economic message: The model separates a country's technological/cost attractiveness from the fixed cost of establishing a sourcing relationship. This distinction is crucial for matching firm-level sourcing patterns.

B	β_c^f	β_d^f	β_f^f	β_c^f	β_{disp}^f
0.122 (0.004)	0.022 (0.002)	0.193 (0.018)	0.872 (0.024)	-0.393 (0.012)	0.934 (0.018)

Notes: Table reports coefficients and standard errors from estimating the model via simulated method of moments. Standard errors based on 25 bootstrap samples drawn with replacement.

Table 5: Estimated Parameters

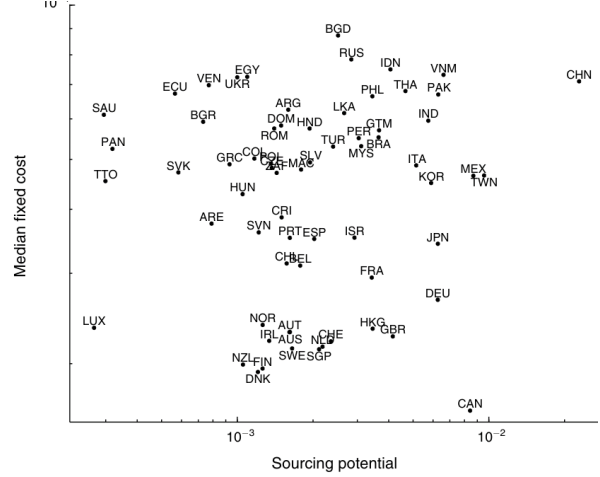


FIGURE 3. ESTIMATED SOURCING POTENTIAL AND MEDIAN FIXED COSTS BY COUNTRY

Figure 3: Estimated Sourcing Potential and Median Fixed Costs by Country

8.5 Figure 4: Model Fit: Share of Firms and Aggregate Sourcing by Country + Table 6: Hierarchies in Sourcing Patterns: Data and Model

Read:

- Figure 4 compares model-predicted and actual country shares of importing firms and aggregate sourcing.
- The model fits the share of importers very closely and also performs well on aggregate import shares.
- Table 6 compares hierarchical sourcing strings in the data and model.
- The model reproduces the overall fraction of firms following the pecking order reasonably well.

Detailed interpretation: Figure 4 evaluates targeted and non-targeted dimensions of fit. Table 6 checks a more subtle pattern: whether firms enter countries in the right order. Matching both country shares and hierarchy patterns is difficult without heterogeneous firm-country fixed costs.

Economic message: The model fits not just aggregate imports, but the structure of firm sourcing choices. This supports using the model for counterfactuals about extensive-margin responses.

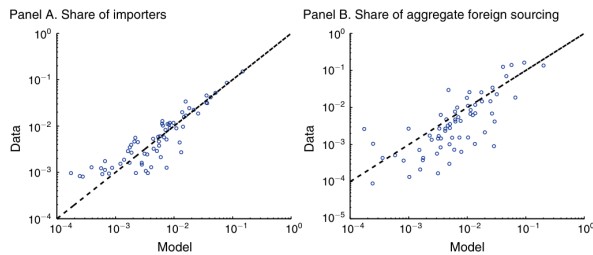


FIGURE 4. MODEL FIT: SHARE OF FIRMS AND AGGREGATE SOURCING BY COUNTRY

Figure 4: Model Fit: Share of Firms and Aggregate Sourcing by Country

TABLE 6—HIERARCHIES IN SOURCING PATTERNS: DATA AND MODEL

String	Data	Baseline model
CA	29.82	28.66
CA-CH	3.67	3.42
CA-CH-DE	0.56	0.58
CA-CH-DE-GB	0.25	0.14
CA-CH-DE-GB-TW	0.13	0.09
CA-CH-DE-GB-TW-IT	0.05	0.02
CA-CH-DE-GB-TW-IT-JP	0.05	0.03
CA-CH-DE-GB-TW-IT-JP-MX	0.08	0.07
CA-CH-DE-GB-TW-IT-JP-MX-FR	0.27	0.12
CA-CH-DE-GB-TW-IT-JP-MX-FR-KR	1.08	0.66
TOTAL following pecking order	36.0	33.8

Notes: This table depicts the percentage of importers following a particular sourcing pattern. The first row shows the percentage of firms only importing from Canada; the second row shows the percentage of firms only importing from Canada and China; and so forth (irrespective of firm sourcing outside these top ten countries). The ranking of countries is determined by the number of firms sourcing from these countries in the data.

Table 6: Hierarchies in Sourcing Patterns: Data and Model

8.6 Table 7: Third Country Sourcing Effects of Chinese Sourcing Potential Shock + Figure 5: Changes in the Size of Firms

Read:

- Table 7 compares the baseline China shock to alternative models.
- In the baseline, new China importers increase domestic sourcing and third-country sourcing.
- Fixed sourcing strategies and independent-entry models mute these responses.
- Figure 5 shows that the China shock generates heterogeneous firm-size effects.
- In the baseline, the largest firms expand while smaller firms contract.

Detailed interpretation: Table 7 is one of the paper’s most important results. It shows that a China shock can raise sourcing from other markets because entrants become larger and more globally active. Figure 5 shows that the same shock reallocates activity across firms: firms able to exploit China sourcing expand, while other firms lose market share.

Economic message: A sourcing shock creates both substitution and productivity/scale effects. The scale effect can dominate for firms that enter China, leading them to increase other sourcing margins too.

TABLE 7—THIRD COUNTRY SOURCING EFFECTS OF CHINESE SOURCING POTENTIAL SHOCK

Chinese import status	Change in sourcing from US	Change in sourcing from other countries	Change in sourcing from China	Share of firms	Share of imports from China
<i>Panel A. Baseline model</i>					
Entrants	1.008	1.015	∞	0.053	0.334
Continuers	1.001	1.001	2.148	0.027	0.666
Others	0.995	0.987	—	0.920	0.000
<i>Panel B. Baseline model, fixed sourcing strategies</i>					
Entrants	—	—	—	0.000	0.000
Continuers	1.002	1.002	2.150	0.027	1.000
Others	0.996	0.996	—	0.973	0.000
<i>Panel C. Independent entry decisions model ($\theta = \sigma - 1 = 2.85$)</i>					
Entrants	0.997	0.994	∞	0.053	0.343
Continuers	0.997	0.996	2.116	0.027	0.657
Others	0.997	0.992	—	0.920	0.000
<i>Panel D. Universal importing model (no fixed costs)</i>					
Entrants	—	—	—	0.000	0.000
Continuers	0.987	0.987	2.976	1.000	1.000
Others	—	—	—	0.000	0.000
<i>Panel E. Common fixed costs across countries model (heterogeneous across firms)</i>					
Entrants	1.006	—	∞	0.108	0.328
Continuers	0.999	0.999	2.249	0.066	0.672
Others	0.993	—	—	0.826	0.000

Notes: The table groups firms by Chinese import status. Entrants are those firms (i.e., bundles of productivity levels and fixed cost draws) that begin sourcing from China. Continuers are firms that source from China before and after the shock. Others are firms that do not source from China before or after the shock. As described in the main text, the shock to the Chinese sourcing potential is calibrated so that the baseline model matches the observed 178 percent increase in the Chinese import share from 1997 to 2007. For the panels C, D, and E, we re-calibrate the shock to the Chinese sourcing potential so that the respective models can match the same 178 percent increase

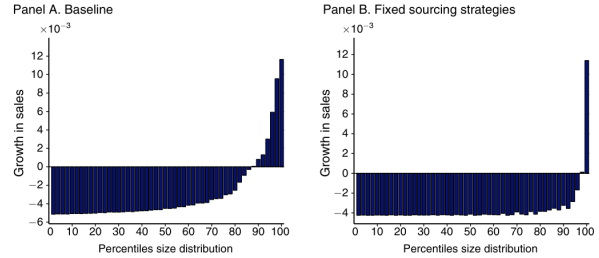


Figure 5: Changes in the Size of Firms

Table 7: Third Country Sourcing Effects of Chinese Sourcing Potential Shock

Figure 5: Changes in the Size of Firms

8.7 Table 8: Gross and Net US Sourcing Effects + Table 9: Observed Changes in Third Country Sourcing from 1997-2007 by Firms’ 2007 Chinese Import Status

Read:

- Table 8 decomposes the gross and net US sourcing effects of the China shock.
- The baseline model predicts both increases and decreases in domestic sourcing across different firms.
- Aggregate domestic sourcing falls, but expanding firms partially offset the decline.
- Table 9 shows actual 1997-2007 sourcing changes by China import status.

- Firms that begin importing from China grow domestic and third-country sourcing the most.

Detailed interpretation: Table 8 shows why aggregate net effects can be misleading: a negative net domestic sourcing change coexists with positive domestic sourcing growth for some firms. Table 9 shows that the qualitative model predictions are visible in the raw data: China entrants expand multiple sourcing margins more than other firms.

Economic message: Interdependencies create gross churning. Some firms replace domestic inputs with Chinese inputs, while others become more productive and buy more inputs from all sources.

TABLE 8—GROSS AND NET US SOURCING EFFECTS

	Baseline		Fixed sourcing strategies	
	Difference in sourcing in billion USD	Change in percent of total US sourcing	Difference in sourcing in billion USD	Change in percent of total US sourcing
Increase in domestic sourcing	4.41	0.11	1.97	0.05
Decrease in domestic sourcing by firms that continue to operate	−13.93	−0.36	−12.86	−0.33
Decrease in domestic sourcing by firms that shut down	−11.11	−0.28	0	0
Total	−20.63	−0.53	−10.88	−0.28

Notes: We use the model's predictions for the US sourcing pre and post the shock to China's sourcing potential in order to calculate percentage changes in US sourcing. We then use aggregate US sourcing purchases in the data and the percentage differences predicted by the model to calculate the predicted USD change in sourcing in response to the shock.

Table 8: Gross and Net US Sourcing Effects

TABLE 9—OBSERVED CHANGES IN THIRD COUNTRY SOURCING FROM 1997–2007 BY FIRMS' 2007 CHINESE IMPORT STATUS

Chinese import status	Change in sourcing from United States	Change in sourcing from other countries	Change in sourcing from China	Share of firms	Share of imports from China
Entrants	2.68	5.93	—	0.08	0.37
Exiters	0.15	0.02	0.00	0.00	0.00
Continuers	1.03	1.32	5.75	0.01	0.63
Others	0.74	1.03	—	0.91	0.00

Notes: This table is based on an unbalanced panel of manufacturing firms from 1997 and 2007. Exiters (entrants) are those firms that stop (begin) sourcing from China. Continuers are firms that source from China in 1997 and 2007. Columns 1, 2, and 3 contain the ratio of total sourcing in 2007 deflated dollars relative to 1997 dollars by each group of firms. The share of firms in column 4 is the number of 2007 firms in each China import status category over total active firms in 2007.

Table 9: Observed Changes in Third Country Sourcing from 1997-2007 by Firms' Chinese Import Status

8.8 Table 10: Estimates of the Impact of the China Shock on Firm-Level Sourcing

Read:

- Table 10 estimates the relationship between firm-level China sourcing shocks and changes in sourcing outcomes.
- OLS and IV estimates generally show that greater China exposure increases domestic inputs, foreign inputs, and the number of source countries.
- The IV results support the idea that the sourcing responses are not only driven by firm-specific demand or productivity shocks.
- Employment effects are less clear, which is consistent with the model focusing on input use rather than firm employment.

Detailed interpretation: Table 10 is the reduced-form counterpart to the structural counterfactual. The key result is that China sourcing exposure is associated with increases in other input-sourcing margins. This is inconsistent with a simple independent-entry or pure substitution model.

Economic message: The reduced-form evidence validates the paper's core mechanism: a positive shock to one source country can expand the firm's entire sourcing network.

TABLE 10—ESTIMATES OF THE IMPACT OF THE CHINA SHOCK ON FIRM-LEVEL SOURCING

Firm-level change from 1997 to 2007 in								
	OLS				IV			
	Domestic inputs	Number of countries	Foreign inputs	Firm empl.	Domestic inputs	Number of countries	Foreign inputs	Firm empl.
China, DHS	0.064 (0.010)	0.255 (0.007)	0.362 (0.013)	0.097 (0.007)	0.758 (0.214)	0.551 (0.080)	0.670 (0.198)	−0.092 (0.162)
Constant	0.054 (0.019)	0.144 (0.013)	0.315 (0.026)	−0.075 (0.014)	−0.054 (0.039)	0.098 (0.017)	0.267 (0.044)	−0.046 (0.032)
Observations	127,400	127,400	127,400	127,400	127,400	127,400	127,400	127,400
First-stage statistics	Coefficient (se) 2.685 (0.505)				KP <i>F</i> -stat 28.25			

Notes: All variables are changes or growth rates from 1997 to 2007. China, DHS is a Davis-Haltiwanger-Schuh growth rate in firm imports from China. Domestic inputs, foreign inputs, and firm employment are a DHS growth rate. Number of countries is the log difference in the number of countries (excluding China, but including the United States) from which the firm sources inputs. Foreign inputs exclude China. Standard errors are in parentheses and clustered by 439 NAICS industries. In the IV specifications, firm-level sourcing from China is instrumented by

Table 10: Estimates of the Impact of the China Shock on Firm-Level Sourcing

9 Detailed result map

9.1 Result block 1: Sourcing decisions are interdependent

The firm's marginal cost depends on the entire set of countries in its sourcing network. Therefore, a country entry decision cannot be evaluated independently of the other countries already in the network.

Why it matters. This breaks the separability that makes canonical export models tractable.

9.2 Result block 2: Complementarity makes the problem solvable

When $(\sigma - 1)/\theta > 1$, source-country entry decisions are complements. Complementarity permits iterative bounds and avoids full enumeration of all country bundles.

Why it matters. This is the bridge from theory to estimation. Without it, the 66-country problem is computationally infeasible.

9.3 Result block 3: Heterogeneous fixed costs are essential

The data show hierarchy but not a strict pecking order. Country-level fixed costs alone cannot match the observed sourcing strings.

Why it matters. Firm-country idiosyncrasy is not a nuisance; it is a central ingredient for matching sourcing patterns.

9.4 Result block 4: China shock effects are heterogeneous

A China sourcing shock expands sourcing margins for some firms and contracts them for others. Aggregate outcomes hide gross firm-level reallocation.

Why it matters. Trade shocks operate through within-industry reallocation among firms with different sourcing networks.

10 Short summary

- The paper builds a model of firm-level global sourcing with many countries and heterogeneous firms.
- Import sourcing differs from exporting because import entry lowers marginal costs.
- The decision to source from one country affects the value of sourcing from other countries.
- Under an empirically relevant parameter condition, source countries are complements.
- Complementarity makes a high-dimensional sourcing problem computationally tractable.
- The model is estimated using confidential US firm-level data on sourcing countries and input purchases.
- Firm-country fixed costs are needed to match the data.
- A firm sourcing from all foreign countries is estimated to have substantially lower variable costs and higher sales.
- A China sourcing shock has heterogeneous effects across firms.
- New China importers expand domestic and third-country sourcing; non-China importers shrink.
- Reduced-form evidence from 1997-2007 supports the model's prediction of positive third-market sourcing responses.

- The paper’s main lesson is that global sourcing shocks reallocate activity through firm-level networks, not simply through aggregate substitution between domestic and foreign inputs.

11 Conclusion

11.1 Core conclusion

Antràs, Fort, and Tintelnot show that the extensive margin of global sourcing is an interdependent firm-level choice. Because importing lowers marginal costs, entering one source country changes the value of entering others. This makes foreign sourcing fundamentally different from standard export-market entry.

11.2 Economic intuition

A firm that gains access to a low-cost country, such as China, becomes more productive and larger. If scale expands enough, the firm may buy more from domestic suppliers and third-country suppliers even while it also imports more from China. The same shock can therefore generate both substitution away from some sources and expansion toward others.

11.3 Incremental contribution to trade economics

The paper’s incremental contribution is to make the extensive margin of importing a first-order object in quantitative trade. It provides a theoretical characterization, a computational algorithm, a structural estimation, and a reduced-form validation of interdependent sourcing choices. It extends the heterogeneous-firm trade tradition beyond exporting and into the multi-country sourcing decisions that define global value chains.

11.4 Empirical contribution

The paper uses confidential US firm-level data to observe global sourcing networks directly. This makes it possible to estimate country sourcing potential, fixed costs, and firm-level sourcing responses rather than inferring them from aggregate trade flows alone.

11.5 Policy and welfare implication

Trade shocks such as China’s integration into global supply chains should not be evaluated only by aggregate import substitution. Sourcing shocks can lower prices, expand some firms, contract others, and change domestic sourcing in both directions. The welfare and employment implications depend on this gross reallocation.

11.6 Limitations

The model abstracts from some supplier-level and input-level details, contractual frictions, and ownership mode choices. Employment effects are also not the central object of the model. These limitations mean the framework should be interpreted as a powerful model of sourcing networks rather than a complete model of all consequences of globalization.

11.7 Takeaway

The central takeaway is that global sourcing is a network decision. The effect of opening one source country depends on every other source in the firm's network, the firm's productivity, and the fixed costs of entering each market. This insight changes how we should understand the firm-level and aggregate effects of globalization.